

BUSINESS ANALYTICS PROJECT

FOOD DELIVERY APPS SENTIMENTAL ANALYSIS DASHBOARD

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Abstract: Understanding Customer Sentiment in Food Delivery Apps: A Power BI Dashboard

This project outlines the development of a Power BI dashboard to analyze customer sentiment towards food delivery apps. By leveraging customer reviews and sentiment analysis, the dashboard provides valuable insights for app companies.

Data: Customer reviews containing text and ratings are analyzed. Sentiment analysis categorizes these reviews as positive, negative, or neutral.

Dashboard Design: Visualizations display overall sentiment, sentiment by app, trending themes, and the relationship between sentiment and ratings. Users can filter by app, timeframe, or keywords for a deeper dive.

Benefits: The dashboard helps app companies understand customer satisfaction, identify areas for improvement, compare competitor performance, and track sentiment over time. This empowers data-driven decision making to enhance customer experience and gain a competitive edge.

Tools: Power BI Desktop facilitates data import, transformation, visualization, and sentiment analysis using built-in tools or external APIs.

Overview:

This project aims to create a user-friendly Power BI dashboard for analyzing customer sentiment towards food delivery apps.

Goal: Gain insights into customer satisfaction and identify areas for improvement within food delivery apps.

Data: We'll utilize customer review data, including text and ratings. Sentiment analysis will categorize these reviews as positive, negative, or neutral.

Key Functionalities:

- **Sentiment Breakdown:** Visualize the overall sentiment distribution across all apps (positive, negative, neutral).
- **App Comparison:** Compare sentiment trends for individual food delivery apps.
- **Thematic Analysis:** Identify key themes through word clouds for positive and negative reviews.
- **Sentiment Trends:** Track sentiment changes over time (monthly, quarterly) to gauge user perception.
- **Review Exploration:** Allow users to drill down and explore sample reviews categorized by sentiment for specific apps.

Benefits:

- **Improved Customer Experience:** Identify areas for improvement based on customer feedback.
- **Data-Driven Decisions:** Track sentiment trends to measure the impact of changes or campaigns.

- **Competitive Advantage:** Benchmark against competitor apps and identify areas for differentiation.

Tools:

- **Power BI Desktop:** The primary platform for building the interactive dashboard with insightful visualizations.
- **Sentiment Analysis:** Leverage Power BI's built-in capabilities or integrate with external APIs

Data Requirements:

The dashboard will require a dataset containing customer reviews or feedback on food delivery apps. This data could include:

Review Text: The actual text of the customer review.

Rating: Numerical rating provided by the customer (e.g., 1-5 stars).

App Name: Name of the specific food delivery app being reviewed.

Date: Date of the review.

Additional Information (Optional): User demographics, order details, etc.

Data Source:

Customer reviews scraped from app stores or review websites.

Internal customer feedback data from the food delivery app itself.

Sentiment Analysis:

You'll need to perform sentiment analysis on the "Review Text" field to categorize reviews as positive, negative, or neutral. This can be done using Power BI's built-in sentiment analysis tools or through external APIs.

Dashboard Design

The dashboard should provide insights into customer sentiment towards various aspects of food delivery apps. Here are some suggested visualizations:

1. Overall Sentiment:

A pie chart or donut chart displaying the distribution of positive, negative, and neutral reviews across all apps.

2. Sentiment by App:

A bar chart comparing the percentage of positive, negative, and neutral reviews for each food delivery app.

3. Word Cloud:

A word cloud highlighting the most frequently used words in positive and negative reviews, allowing for quick identification of key themes.

4. Sentiment Over Time:

A line chart showing how sentiment towards each app changes over time (monthly, quarterly).

5. Reviews by Rating:

A scatter plot or table visualizing the relationship between sentiment score and numerical rating provided by the customer.

6. Drill-Down Functionality:

Allow users to drill down into specific apps and view sample reviews categorized by sentiment.

Additional Considerations

Filters: Include slicers or filters to allow users to focus on specific apps, timeframes, or keywords.

KPIs: Consider displaying key performance indicators (KPIs) such as average rating or percentage of positive reviews.

Customization: Allow users to customize the dashboard by hiding or rearranging visualizations.

Tools and Techniques

Power BI Desktop: This is the primary tool for building the dashboard.

Data Import and Transformation: Power Query Editor can be used to clean and transform the data.

Sentiment Analysis: Utilize Power BI's built-in sentiment analysis or integrate with external sentiment analysis APIs.

Data Visualization: Utilize Power BI's extensive library of charts and graphs to create insightful visualizations.

Benefits

This dashboard will provide valuable insights for food delivery app companies, such as:

- * Understanding customer satisfaction with different aspects of the service (delivery speed, restaurant selection, customer support).
- * Identifying areas for improvement based on negative sentiment.
- * Comparing performance against competitor apps.
- * Tracking sentiment over time to measure the impact of changes or campaigns.

By leveraging customer feedback through sentiment analysis, food delivery apps can improve customer experience and gain a competitive edge.

Future Enhancement:

- Integration with social media data to capture a broader spectrum of customer sentiment.
- Sentiment analysis by category (e.g., delivery speed, restaurant selection) for more granular insights.
- Advanced machine learning techniques for predicting future sentiment trends.

Conclusion:

This project successfully developed a comprehensive Power BI dashboard for analyzing customer sentiment towards food delivery apps. By harnessing the power of sentiment analysis on customer reviews, the dashboard equips food delivery companies with valuable insights to improve their services.

Screenshots:

